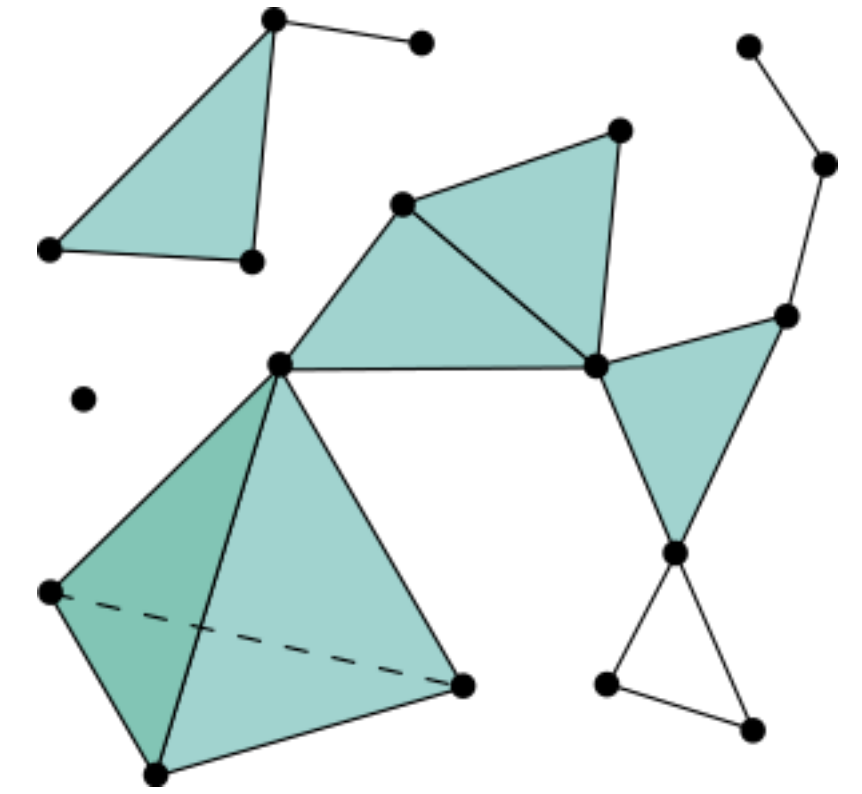


Learning on Simplicial Complexes: Potentials in biomolecular field



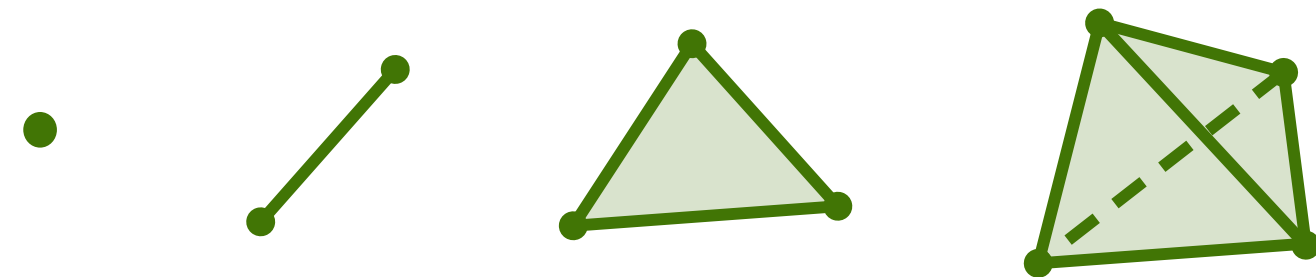
- **simplicial complexes: a network model (richer topological information)**
- **learning: convolution, NNs, probabilistic models, generative models**
- **lies in the field of topological (and geometric) deep learning**

Maosheng Yang

Collaborators: Elvin Isufi (TU Delft), Geert Leus (TU Delft),
Viacheslav Borovitskiy (ETH Zurich),
Michael T. Schaub (Aachen)

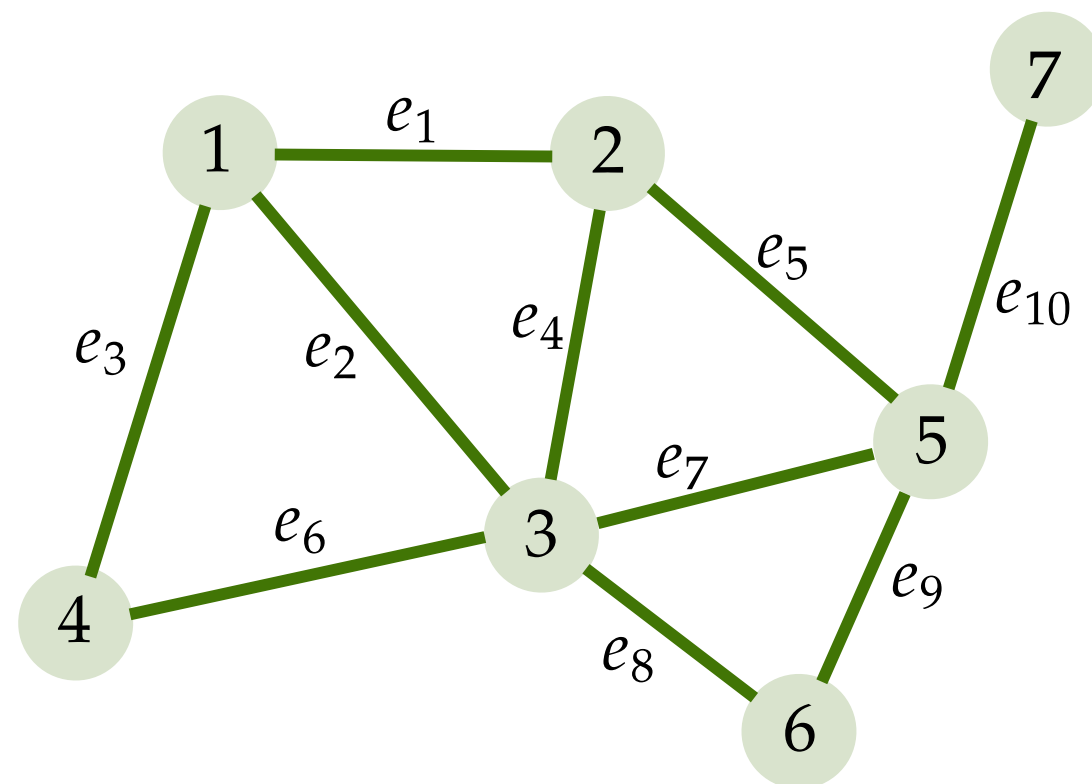
Graphs vs Simplicial 2-Complexes

- Where are SCs used?
- Network analysis
 - Topological data analysis
 - Geometric data processing
 - Computational physics
 - Computer graphics
 - ...
 - Topological deep learning
 - Geometric deep learning

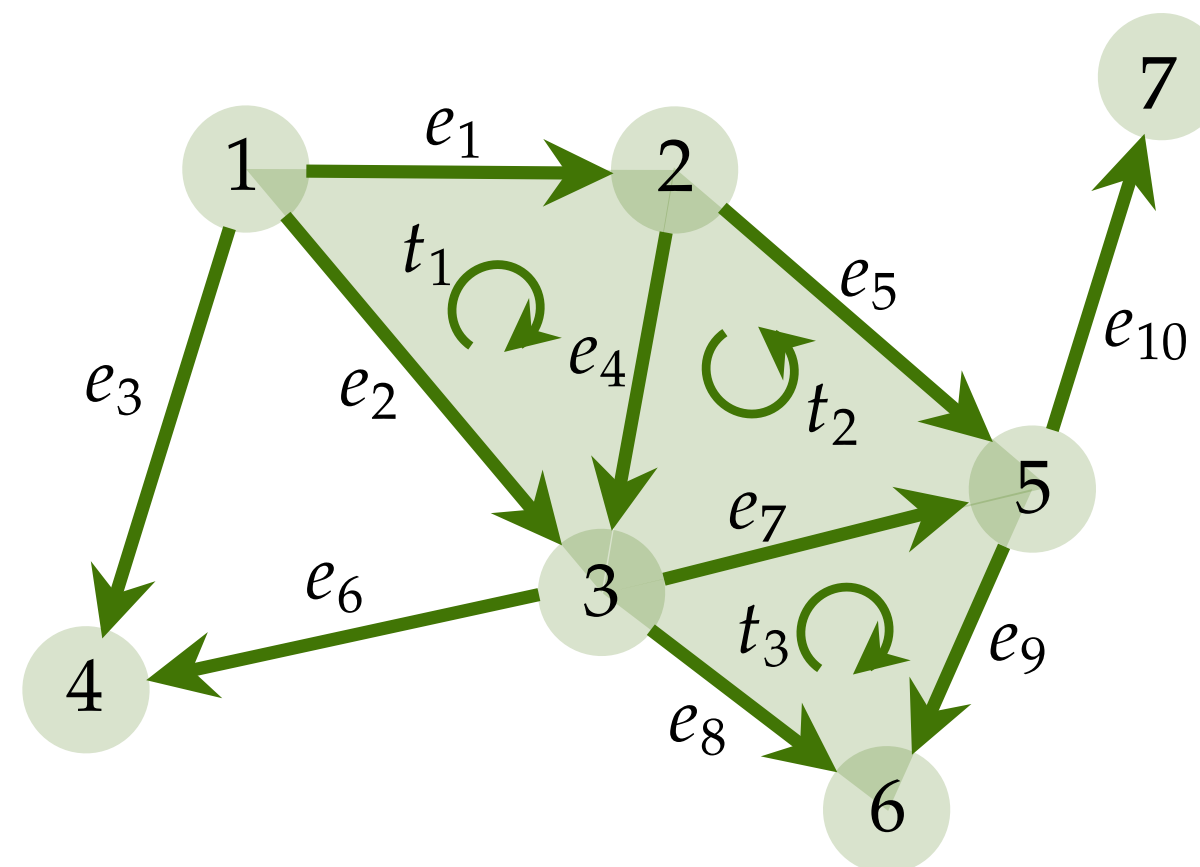


0-, 1-, 2-, 3-simplices

node edge triangle tetrahedron

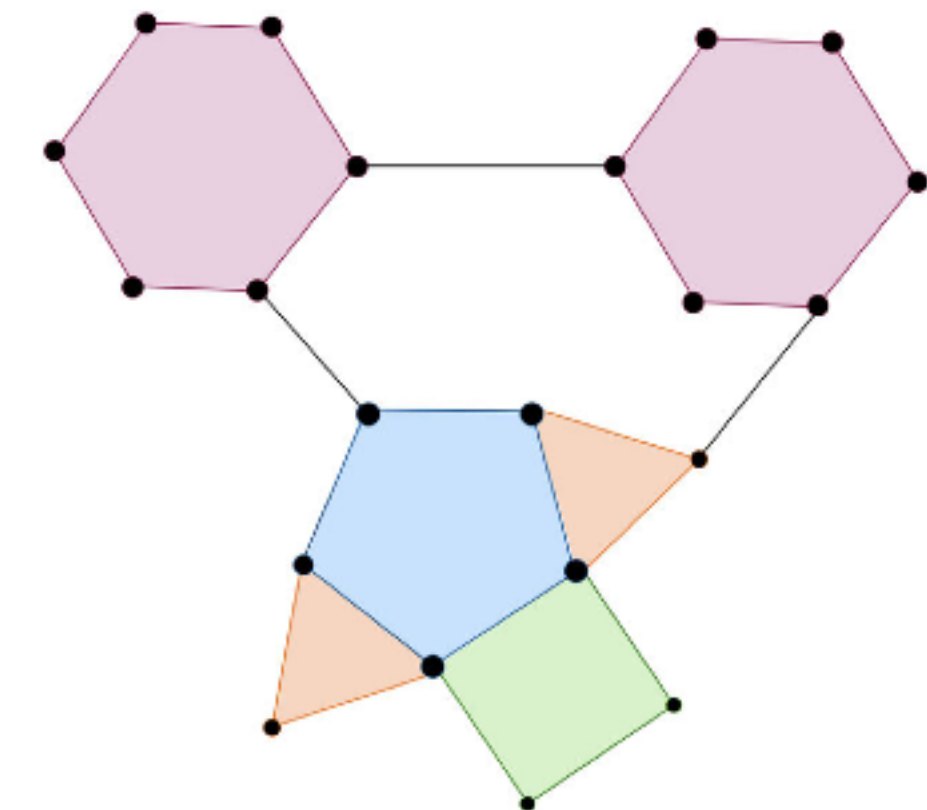


Graph = Simplicial 1-complex



Simplicial 2-complex

- Give simplices orientations



Cell complex

- To model Higher-order network structure
- To support Higher-order signals